COSC 4370 - Homework 1

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1 Objective

The goal of this assignment is to implement an algorithm for rasterizing an ellipse. Specifically, rasterizing the ellipse defined by: $(\frac{x}{16})^2 + (\frac{y}{8})^2 = 45^2$ where x ≥ 0. Modifying this equation to its standard form we get: $(\frac{x}{720})^2 + (\frac{y}{360})^2 = 1$, where we can see a = 720, and b = 360. Taking these values into consideration, the dimension of the image will be 1800 x 900 (2.5a x 2.5b).

2 Method

To accomplish this objective, the function *rasterizeEllipse* was created. It takes 3 parameters: a, b, and a reference to a BMP object. Similar to the midpoint circle algorithm, which only operates over a single octant of a circle, the function operates on two regions of the ellipse, and uses symmetry to rasterize the regions symmetrical to the x-axis. Due to the required specifications, regions located in the -x direction were not rasterized, though commented out code does exist which would account for such regions.

3 Implementation Details

Much like the circle algorithm, this algorithm begins at the northmost point (0, b) and continues clockwise, using the decision variable sigma to choose either the Eastern pixel or the Southeastern pixel.

The mathematical function used for this algorithm is λ_i = F(x, y) = $b^2x^2 + a^2y^2 - a^2b^2$, where (x, y) is the pixel last drawn. Beginning at point (0, b), the decision variable sigma would be calculated as $\alpha = \lambda$(x, y) + λ(x, y – 1) = $2b^2 + a^2(1 - 2b)$.

After each iteration, if $\alpha \leq 0$, then the East choice is selected. Else, α is incremented by $4a^2(1 - y)$, and y is decremented; this is the Southeast choice. Regardless of its state, it is then incremented by $2b^2(2x + 3)$, and the process continues again.

This would continue until the end of the region, at which point the inequality $b^2x \leq a^2$ becomes false. Then, the second region would begin at point (a, 0) and continue a similar process, this time operating counterclockwise.

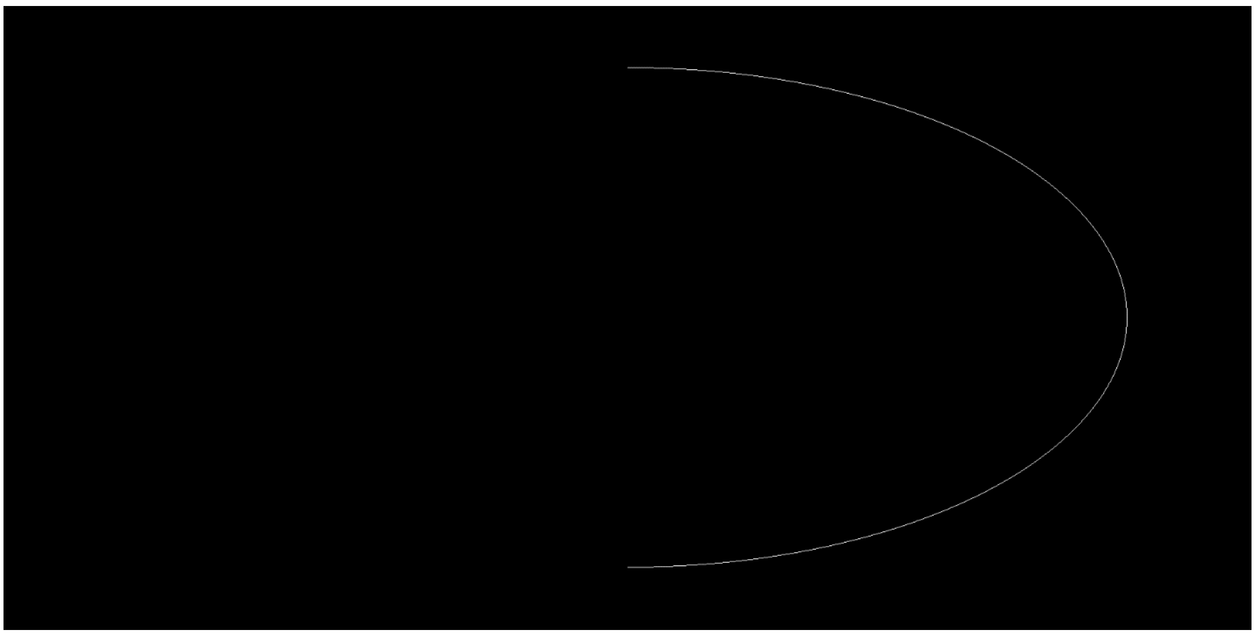
Once a pixel has been selected, we use the provided BMP object to draw it as well as its symmetric counterpart.

In practice, the majority of the incremental values are stored as variables, and a loop exists for each region. The loops themselves are almost entirely identical, simply swapping x variables with y and a variables with b.

Additionally, since our image dimensions are 1800x900, each pixel is transformed 900 units to the positive x direction, and 450 units to the positive y direction to be centered in this image plane.

4 Results

The output produced from this program is a BMP file. Seen in the file, is a white ellipse on a black background.



5 Other Materials

This link directs to a greatly referred to algorithm for ellipse rasterization used for this homework.

https://www.geometrictools.com/Documentation/IntegerBasedEllipseDrawing.pdf